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PALEONTOLOGY

# Athletic capability index in pterosaurs: an initial assessment

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**Abstract:** Pterosaurs had a unique morphology among flying vertebrates, and therefore, a direct comparison with living vertebrates regarding their terrestrial locomotion is impossible. In this study, we used the athletic capability indexes to infer aspects of the terrestrial proficiency of pterosaurs. We estimated the body mass, bending strength, and stylopodial axial strength indexes of 25 specimens of non-pterodactyloid and pterodactyloid pterosaurs. We used linear regression models to investigate the relationships between the indexes of athletic capability and the two stylopodia, and the indexes of athletic capability and the body mass of pterosaurs. We also compared the values of the index with those of other extinct and living animals. The results show higher values of the indexes of athletic capability for the humerus than the femur, indicating a higher relative strength of the forelimbs compared to the hindlimbs for bending and axial compression forces. We also detected a negative relationship between both indexes of athletic capability and body mass. Comparing the pterosaur athletic capability with other quadruped vertebrates indicates they had a good proficiency for terrestrial locomotion, even for plesiomorphic forms previously thought to have limited terrestrial locomotion. Even so, pteranodontoids are found to exhibit lower femoral ACI if compared to similar-sized pterodactyloids, notably azhdarchoids and dsungaripterids.

**Key words:** Biomechanics, Quadrupedal locomotion, Bone strength, Pterosauria.

## INTRODUCTION

Pterosauria constitutes a diverse clade of archosaurian reptiles that lived during the Mesozoic Era, appearing in the fossil record at the end of the Triassic period and disappearing in the Late Cretaceous extinction event. They were the first known winged vertebrates capable of active flight, having forelimbs with a hyper-elongated fourth finger that supported the wing, which consisted of a membrane extending along the body to the hindlimb (Kellner et al. 2009, Palmer 2010, Elgin et al. 2011). This structural conformation allowed pterosaurs to develop a peculiar way of walking, using all four limbs (Stokes 1957, Unwin 1988, Chatterjee & Templin 2004, Witton 2008, 2015, Mazin et al. 2009, Costa et al. 2013, Chan 2017). Pterodactyloid tracks reveal that handprints appear deeper in the sediment than footprints, indicating that most of the animal's weight was borne by the forelimbs on the ground (Hwang et al. 2002, Chatterjee & Templin 2004). This historical evidence highlights that some structural variations of pterosaurs are likely associated with terrestrial locomotion functions and not just aspects of the active flight (Witton & Naish 2008, Chan 2017).

Despite many studies exploring pterosaur flight (e.g., Bennett 2000, 2008, Chatterjee & Templin 2004, Habib 2008, Palmer 2010, Witton & Habib 2010, Naish et al. 2021, Padian et al. 2021, Griffin et al. 2022, 2024; for revisions, see Middleton & English 2015, Martin-Silverstone et

al. 2020), their terrestrial locomotion is relatively less studied. Though pterosaur stance and gait were once heatedly debated, it is now widely accepted that they were apt quadrupeds (e.g., Lockley et al. 1995, Mazin et al. 1995, 2003, Unwin 1996, Bennett 1997, Fastnacht 2005, Witton & Naish 2008, Costa et al. 2013), moving with erect hindlimbs and a parasagittal stride (e.g., Mazin et al. 1995, 2003, Costa et al. 2013). Also, there is evidence of distinct terrestrial locomotion between pterosaur groups, which remains open for examination (see Witton 2015, Mazin & Pouech 2020). For example, there is a possibility that some pterodactyloids, notably members of Azhdarchoidea, were even more adapted to walking on land than other pterosaurs (Witton & Naish 2008). In this case, some pterodactyloids would act similarly to birds that specialize on feeding on small prey on land or shallow water bodies, such as herons and shoebills, a hypothesis based on the robustness of their forelimbs (Witton & Naish 2008, Naish & Witton 2017, Padian et al. 2021).

Terrestrial locomotion imposes forces and stresses on bones, muscles, and joints that vary depending on the activity and where vertebrates perform such activity: if animals are standing still, walking, running, or even about to take off, in the case of a flying vertebrate. Also, body mass is a crucial factor in determining the ability of different vertebrate species to move (Alexander 1983, 1985, Selker & Carter 1989, Farlow et al. 1995, Campione & Evans 2012). Understanding whether a vertebrate's bones can withstand a given stress force is vital to inferring its ability to move with less or more resourcefulness in a terrestrial environment (Alexander 1985, 1989, Fariña 1995, 2001, Fariña et al. 2005). In this respect, a vertebrate with greater proficiency in terrestrial locomotion is expected to have relatively more resistant appendicular bones than an animal with less proficiency (Alexander 1985, 1989, Christiansen 1998, Sarin et al. 1999, Fariña 2001). Therefore, it is possible to infer the pterosaur functionalities of long bones by deepening the comparative studies of body size, bone resistance, and biomechanics across different groups.

An efficient biomechanical method to determine the resistance of the long bones of the limbs is calculating the Athletic Capability Index (ACI) (Alexander 1985, 1989, Sarin et al. 1999, Fariña 1995, 2001, Fariña et al. 2005). Using this approach, one can infer the bending and axial forces that the humerus and femur would be capable of withstanding, taking into account their strength and the estimated mass of the animal (Casinos 1996, Christiansen 1998, Fariña 2001). This approach has already been applied to dinosaurs (e.g. Alexander 1989, Christiansen 1998), mammals (Casinos 1996, Fariña 1995, 2001, Fariña et al. 2005, Rocha-Barbosa & Casinos 2011), and some extant birds (Alexander 1989, Casinos 1996). Key biomechanical studies have already explored pterosaur long bone strength indicators, including section modulus (Witton & Habib 2010, Naish et al. 2021) and relative failure force (Naish et al. 2021). Still, the ACI itself, which also takes body mass into account, has not yet been estimated for pterosaurs.

In this study, our main objective is to investigate the athletic capability of the humerus and femur, the body mass, and their relationship to characterize the terrestrial locomotion capabilities of pterosaurs. For this purpose, we recovered a data set of 25 specimens, encompassing pterodactyloid and non-pterodactyloid species, ranging in estimated body mass from 0.02 to +200 kg. Characterizing the athletic capability of different pterosaurs may reveal the group's diversity of terrestrial locomotion capabilities, contributing to evaluating the posture and terrestrial proficiency in pterosaurs.

## MATERIALS AND METHODS

### Institutional abbreviations

AMNH, American Museum of Natural History, New York, EUA; BSPG, Bayerische Staatssammlung fur Palaontologie und historische Geologie, Munich, Germany; FGGUB, Faculty of Geology and Geophysics of the University of Bucharest, Bucharest, Romania; GP/2E, Laboratório de Paleontologia Sistemática do Instituto de Geociências da Universidade de São Paulo, São Paulo, Brazil; IVPP, Institute of Vertebrate Paleontology and Paleoanthropology, Chinese Academy of Sciences, Beijing, China; MGUH, Geological Museum of Copenhagen, University of Copenhagen, Copenhagen, Denmark; MHIN-UNSL-GEO-V, Museo de Historia Natural de la Universidad Nacional de San Luis, San Luis, Argentina; MN, Museu Nacional/Universidade Federal do Rio de Janeiro, Rio de Janeiro, Brazil; MOR, Museum of the Rockies, Bozeman, EUA; NHMUK, Natural History Museum, London, United Kingdom; NSM, National Science Museum, Tokyo, Japan; RAM, Raymond M. Alf Museum of Paleontology, Claremont, EUA; RBCM, Royal British Columbia Museum, Victoria, Canada; RGM, Nationaal Natuurhistorisch Museum (Naturalis), Leiden, Netherlands; SHN, Sociedade de História Natural, Torres Vedras, Portugal; SMNK, Staatliches Museum fur Naturkunde, Karlsruhe, Germany; SMNS, Staatliches Museum fur Naturkunde, Stuttgart, Germany; TMM, Texas Memorial Museum, Austin, USA; TMP, Royal Tyrrell Museum of Palaeontology, Drumheller, Canada; USNM, National Museum of Natural History, Washington, D.C., USA; YPM, Peabody Museum of Natural History, Yale University, New Haven, USA.

### Analyzed specimens

To characterize the terrestrial locomotion capabilities of pterosaurs, we first selected measurements of the humerus and femur lengths and cortical tissue thickness at the bone diaphysis as available in the literature, or else images of the transverse section of such bones with the potential of measuring these features. We thus selected a dataset of 25 specimens of pterosaurs, covering non-pterodactyloid and pterodactyloid species (Table I). This selection covers a relatively broad spectrum of pterosaur phylogenetic diversity. When there was no corresponding femur or humerus in the same individual, the values were estimated (scaled) from known fossils of other specimens of the same species or the phylogenetically closest relatives available, corrected to the same body size (as per humeral/femoral ratios seen in the same species or phylogenetically closer relatives). Table SI – Supplementary Material shows which values were taken directly from the literature or estimated from other specimens/relatives (see Table SI for further details).

### Athletic Capability estimation

The Athletic Capability Index (*ACI*) is represented by two strength indicators, (Alexander 1985, 1989, Casino 1996, Fariña 1995, 2001, Fariña et al. 2005), the Bending Strength Indicator *(BSI*) and the Axial Strength indicator (*ASI*).

The BSI is expressed by the equation:

$$BSI = Z/(a.m.g.x)$$

When the stylopodial bone (humerus or femur) is subjected to a perpendicular force exerted on its long axis, its resistance to bending is proportional to the section modulus (*Z),* a property of its diaphysis cross-section, and inverse of the percentage (*a*) of the mass withstood by the limb pair (fore or hind), the body mass (*m*), the gravitational coefficient (*g*), and the straight-line distance (*x*) from the measured cross-section to the distal joint of the bone (Alexander 1985, 1989, Fariña 2001, Morato 2006, Saffar et al. 2009).

**Table I. The 25 studied specimens.**

| Group | Species/specimens | Main sources |
|---|---|---|
| Eudimorphodontoidea | *Arcticodactylus cromptonellus* (MGUH VP 3393) | Jenkins et al. 2001 |
| Rhamphorhynchidae | *Rhamphorhynchus muensteri* (RAM 14522) | Naish et al. 2021 |
| Rhamphorhynchidae | *Rhamphorhynchus muensteri* (SMNS 9620) | Naish et al. 2021 |
| Rhamphorhynchidae | Rhamphorhynchidae indet. (NHMUK PV R 40126) | Naish et al. 2021 |
| Anurognathidae | *Anurognathus ammoni* (Private specimen Bennett 2007) | Bennett 2007, Naish et al. 2021 |
| Germanodactylidae | *Germanodactylus ramphastinus* (BSPG 1977 XIX1) | Naish et al. 2021 |
| Ctenochasmatidae | *Pterodaustro guinazui* (MHIN-UNSL-GEO-V-241, hatchling) | Codorniú & Chiappe 2004, Naish et al. 2021 |
| Anhangueridae | *Anhanguera piscator* (NSM-PV 19892) | Kellner & Tomida 2000, Naish et al. 2021 |
| Anhangueridae | Anhangueridae indet. (BSP 1980 I 122) | Naish et al. 2021 |
| Anhangueridae | Anhangueridae indet. (SMNK 1133) | Naish et al. 2021 |
| Anhangueridae | "*Brasileodactylus* sp." (AMNH 24444) | Naish et al. 2021 |
| Anhangueridae | *Anhanguera spielbergi* (RGM 401 880) | Veldmeijer 2003 |
| Anhangueridae | *Tropeognathus mesembrinus* (MN 6594-V) | Kellner et al. 2013 |
| Pteranodontidae | *Pteranodon longiceps* (YPM 1175) | Eaton 1910 |
| Dsungaripteridae | Dsungaripteridae indet. (SHN.13) | Bertozzo et al. 2021 |
| Dsungaripteridae | *Noripterus complicidens* (IVPP RV 73001) | Jiang et al. 2023, Song et al. 2024 |
| Azhdarchoidea indet. | *Bennettazhia oregonensis* (USNM 11925) | Naish et al. 2021 |
| Tapejaridae | "*Nemicolopterus crypticus*" (IVPP V-14377, Sinopterinae indet., hatchling) | Wang et al. 2008, Naish et al. 2021 |
| Tapejaridae | *Tupandactylus navigans* (GP/2E 9266) | Beccari et al. 2021 |
| Tapejaridae | *Tapejara wellnhoferi* (SMNK PAL 3985) | Elgin & Hone 2013 |
| Azhdarchidae | *Cryodrakon boreas* (TMP 1992.83) | Hone et al. 2019 |
| Azhdarchidae | *Hatzegopteryx thambema* (FGGUB R1083) | Naish et al. 2021 |
| Azhdarchidae | *Quetzalcoatlus lawsoni* (TMM 47180) | Hone et al. 2019 |
| Azhdarchiformes | *Montanazhdarcho minor* (MOR 691) | Naish et al. 2021 |
| Azhdarchoidea indet. | Azhdarchidae? indet. (RBCM. EH.2009.019.0001) | Naish et al. 2021 |

The section modulus describes the resistance of the cross-section of a given material. The higher the section modulus, the more resistant the section (Sarin et al. 1999, Ivanov 2008). We calculate the section modulus of the cross-section of the stylopodial diaphysis, the bone part most subject to mechanical stress during terrestrial locomotion (Alexander 1989, Currey 1999, Fariña 1995, 2001, Elissamburu 2007, Rocha-Barbosa & Casinos 2011). Here, we approximate the diaphyseal section as a (hollow) circle, as in some previous studies (e.g. Witton 2008, Naish et al. 2021). Although pterosaur stylopodials have a roughly oval (almost circular) shape with differences in sagittal and transverse diameters, such measurements are unavailable

for most specimens. Adopting the hollow circle simplification does not significantly affect the section modulus estimation (Saffar et al. 2009, Naish et al. 2021). An average value for the diaphyseal diameter was used for fossils whose sagittal and transverse diameter information was available, such as for *Cryodrakon boreas*. The section modulus is expressed using the following equation:

$$Z = (\pi.(D^4-d^4))/32D$$

Following the equation, *D* is the total diameter of the diaphysis, and *d* is the inner diameter of the diaphysis. The inner diameter can be obtained by subtracting the cortical tissue thickness from the total diameter of the diaphysis. Cortical tissue thickness values were obtained from the literature. Without this information for a given species, we got a cortical tissue thickness estimate scaled from the phylogenetically closest species available (for further details, see the Supplemental Information for an Excel file comprising the data on calculations and estimations). A gravitational constant of $9.8m/^2$ was adopted. The value of the *x* was determined by measuring the region of the bone diaphysis where the cross section was measured up to the distal portion of the stylopodium.

For extinct animals, the average weight that each limb can support when walking can be estimated using physical scale models (Alexander 1985, 1989, Fariña 2001) or from the position of the centre of mass in relation to the limbs (Fariña 2001, Morato 2006). The fraction of weight that each limb supports is inversely proportional to the distance of its point of support from the projection of the centre of mass on the surface (Alexander 1989, Fariña 2001, Morato 2006).

For the present study, pterosaur centres of mass were taken from the work of Henderson (2024), while the distances between each limb's point of support and the projection of the centre of mass on the surface were taken from reference standing postures as reconstructed by Witton (2013, 2015). For an example, see Figure 1. See Table II for further details on data sources for each analyzed specimen (choice of source was based on phylogenetic/morphological proximity).

The ASI is expressed by the equation:

$$ASI = A/(m.g.a)$$

In this case, the resistance of the bone to axial stresses is proportional to the cross-sectional area of its diaphysis (*A*), the inverse of the body mass (*m*) multiplied by the gravitational coefficient (*g*) and the percentage of weight that each limb supports (*a*). The diaphysis solid cross-sectional area (*A*) was expressed in $mm^2$, and calculated by subtracting the area of the medullary cavity (considering *r* as the inner diameter, half of *d*) from the area of the shaft section (considering *R* as the outer diameter, half of *D*):

$$A = (\pi*R^2) - (\pi*r^2)$$

In quadrupedal vertebrates, the ACI of each stylopodial should be measured separately: the humerus ($ACI_h$) and the femur ($ACI_f$). For bipedal animals, the ACI is calculated only for the femur since the femur supports 100% of the animal's mass, excluding the variable *a* from the equation (see Alexander 1985 and Christiansen 1998). For both ACI, the resulting values are expressed in $Pa^{-1}$ (or $m^2N^{-1}$).

**Table II.** The 25 studied specimens and their respective estimated values of anterior and posterior *a*.

| Group | Species | a (anterior) | a (posterior) | Reference for position of center of mass | Reference standing pose |
|---|---|---|---|---|---|
| Eudimorphodontoidea | *Arcticodactylus cromptonellus* (MGUH VP 3393) | 0,53 | 0,47 | Henderson (2024), *Eudimorphodon* | Witton (2015), *Preondactylus* |
| Rhamphorhynchidae | *Rhamphorhynchus muensteri* (RAM 14522) | 0,6 | 0,4 | Henderson (2024), *Rhamphorhynchus* | Witton (2015), *Rhamphorhynchus* |
| Rhamphorhynchidae | *Rhamphorhynchus muensteri* (SMNS 9620) | 0,6 | 0,4 | Henderson (2024), *Rhamphorhynchus* | Witton (2015), *Rhamphorhynchus* |
| Rhamphorhynchidae | Rhamphorhynchidae indet. (NHMUK PV R 40126) | 0,6 | 0,4 | Henderson (2024), *Rhamphorhynchus* | Witton (2015), *Rhamphorhynchus* |
| Anurognathidae | *Anurognathus ammoni* (Private specimen Bennett 2007) | 0,52 | 0,48 | Henderson (2024), *Anurognathus* | Witton (2015), *Anurognathus* |
| Germanodactylidae | *Germanodactylus ramphastinus* (BSPG 1977 XIX1) | 0,6 | 0,4 | Henderson (2024), *Germanodactylus* | Witton (2013), *Germanodactylus* |
| Ctenochasmatidae | *Pterodaustro guinazui* (MHIN-UNSL-GEO-V-241, hatchling) | 0,54 | 0,46 | Henderson (2024), *Pterodaustro* | Witton (2013), *Pterodaustro* |
| Anhangueridae | *Anhanguera piscator* (NSM-PV 19892) | 0,45 | 0,55 | Henderson (2024), *Anhanguera* | Witton (2013), *Tropeognathus* |
| Anhangueridae | Anhangueridae indet. (BSP 1980 I 122) | 0,45 | 0,55 | Henderson (2024), *Anhanguera* | Witton (2013), *Tropeognathus* |
| Anhangueridae | Anhangueridae indet. (SMNK 1133) | 0,45 | 0,55 | Henderson (2024), *Anhanguera* | Witton (2013), *Tropeognathus* |
| Anhangueridae | "*Brasileodactylus* sp." (AMNH 24444) | 0,45 | 0,55 | Henderson (2024), *Anhanguera* | Witton (2013), *Tropeognathus* |
| Anhangueridae | *Anhanguera spielbergi* (RGM 401 880) | 0,45 | 0,55 | Henderson (2024), *Anhanguera* | Witton (2013), *Tropeognathus* |
| Anhangueridae | *Tropeognathus mesembrinus* (MN 6594-V) | 0,45 | 0,55 | Henderson (2024), *Anhanguera* | Witton (2013), *Tropeognathus* |
| Pteranodontidae | *Pteranodon longiceps* (YPM 1175) | 0,4 | 0,6 | Henderson (2024), *Pteranodon* | Witton (2013), *Pteranodon* |
| Dsungaripteridae | Dsungaripteridae indet. (SHN.13) | 0,5 | 0,5 | Henderson (2024), *Dsungaripterus* | Witton (2013), *Noripterus* |
| Dsungaripteridae | *Noripterus complicidens* (IVPP RV 73001) | 0,5 | 0,5 | Henderson (2024), *Dsungaripterus* | Witton (2013), *Noripterus* |
| Azhdarchoidea indet. | *Bennettazhia oregonensis* (USNM 11925) | 0,51 | 0,49 | Henderson (2024), *Tupuxuara* | Witton (2013), *Huaxiadraco* |
| Tapejaridae | "*Nemicolopterus crypticus*" (IVPP V-14377, Sinopterinae indet., hatchling) | 0,51 | 0,49 | Henderson (2024), *Tupuxuara* | Witton (2013), *Huaxiadraco* |
| Tapejaridae | *Tupandactylus navigans* (GP/2E 9266) | 0,51 | 0,49 | Henderson (2024), *Tupuxuara* | Witton (2013), *Huaxiadraco* |
| Tapejaridae | *Tapejara wellnhoferi* (SMNK PAL 3985) | 0,51 | 0,49 | Henderson (2024), *Tupuxuara* | Witton (2013), *Huaxiadraco* |
| Azhdarchidae | *Cryodrakon boreas* (TMP 1992.83) | 0,57 | 0,44 | Henderson (2024), *Quetzalcoatlus* | Witton (2013), *Zhejiangopterus* |
| Azhdarchidae | *Hatzegopteryx thambema* (FGGUB R1083) | 0,57 | 0,44 | Henderson (2024), *Quetzalcoatlus* | Witton (2013), *Zhejiangopterus* |
| Azhdarchidae | *Quetzalcoatlus lawsoni* (TMM 47180) | 0,57 | 0,44 | Henderson (2024), *Quetzalcoatlus* | Witton (2013), *Zhejiangopterus* |
| Azhdarchiformes | *Montanazhdarcho minor* (MOR 691) | 0,57 | 0,44 | Henderson (2024), *Quetzalcoatlus* | Witton (2013), *Zhejiangopterus* |
| Azhdarchoidea indet. | Azhdarchidae? indet. (RBCM. EH.2009.019.0001) | 0,57 | 0,44 | Henderson (2024), *Quetzalcoatlus* | Witton (2013), *Zhejiangopterus* |

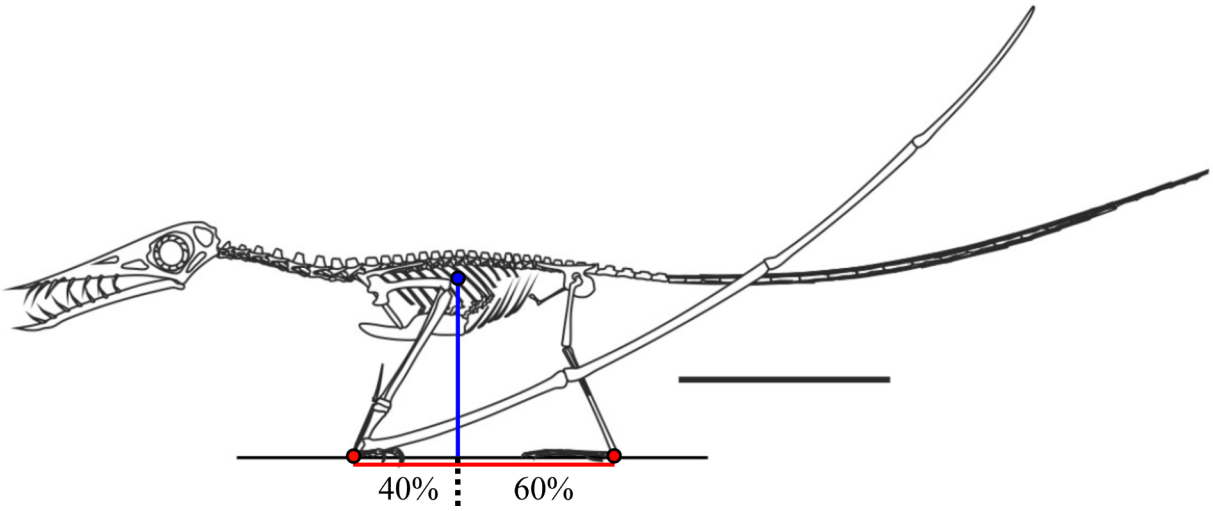



**Figure 1. Weigth limb support in the pterosaur *Rhamphorhynchus muensteri*, as reconstructed in standing pose (Witton 2015). The blue circle indicates the center of mass (after Henderson 2024). The red circles indicate the support point for each limb (fore and hind limb). Weight support for each limb is inversely proportional to the distance between their support point and the perpendicular projection, on the ground, of the center of mass. In the depicted example, weight support for fore and hindlimbs is thus 60% and 40%, respectively. *Rhamphorhynchus muensteri* artwork modified from Witton (2015), published under a CC BY 4.0 license.**

## Body mass estimation

The body mass of pterosaurs is a key factor determining their ability to move in terrestrial environments (Chatterjee & Templin 2004, Witton 2015). However, body mass estimation in pterosaurs is a subject of intense debate, and several methods have been used to estimate it (Witton 2008, Henderson 2010, Witton & Habib 2010, Silverstone & Palmer 2014, Pêgas et al. 2021). Here, we calculated the body mass using the same method for each analyzed specimen, to avoid disparate estimations based on distinct methods (see Witton 2008). We applied the wingspan/body mass correlation provided by Witton (2008), which is based on a body masses as estimated from skeletal volume (Witton 2008). We chose this method because it allows calculating the body mass of a vast array of pterosaurs in a forward manner, requiring only a wingspan estimate. As our dataset already includes humerus length values, wingspan estimates could be promptly made even for fragmentary specimens (based on humerus length/wingspan proportions of closely related taxa; see Table SI for further details). The correlations for non-pterodactyloids and pterodactyloids are as follows:

$$M = 0.681w^{2.807} \text{ (for non-pterodactyloids)}$$

$$M = 0.519w^{2.55} \text{ (for pterodactyloids)}$$

In this case, *M* denotes the body mass (*M*) while *w* denotes wingspan.

## Statistical analyses

To investigate differences in the bending strength indicator (*BSI*) and the axial strength indicator (*ASI*) between long bones (humerus and femur), we constructed two linear models (LMs) with Gaussian distribution, in which the response variable was the BSI and ASI, and the categorical predictor was the long bones. Since the distribution of residuals did not follow the assumptions of homogeneity of variances, we applied the log transformation to the variables

before analyses. However, we kept the original variables in the graphs.

To investigate the relationship between the bending strength indicator (*BSI*) and the axial strength indicator (*ASI*) with the pterosaur body mass, we constructed four linear models (LMs) with Gaussian distribution, in which the response variable was the *BSI* (humerus or femur) and *ASI* (humerus or femur). The continuous predictor was body mass. The analyses and corresponding graphs treated the humerus and femur indexes separately. Since relationships between these indexes and body mass are nonlinear (see Fariña et al. 1997, Rocha-Barbosa & Casinos 2011), we log-transformed response variables and predictors before all analyses.

## RESULTS

The body mass ranged from 0.02 kg to 366 kg across the 25 pterosaur taxa (Table III). The bending strength indicator (*BSI)* values showed a range of 17 to 461 $MPa^{-1}$ for the humeri and 12 to 240 $MPa^{-1}$ for the femora. The axial strength indicator (*ASI)* values showed a range of 0.2 to 9.4 $GPa^{-1}$ for the humeri and 0.5 to 9.9 $GPa^{-1}$ for the femora. These values are summarized in Table III.

The BSI and ASI showed distinct patterns across bones (Figure 2a, b). For BSI, the humeri had higher mean values than the femora ($F(1,48) = 5.9$, $p = 0.018$), indicating greater resistance to bending forces. In contrast, mean ASI values did not differ significantly between bones ($F(1,48) = 0.31$, $p = 0.577$), suggesting similar resistance to axial compression. There were negative relationships between the *BSI* and *ASI* indexes with the body mass of pterosaurs, independent of the long bone considered – Figure 2C and D. On average, we detected a large effect size of body mass on the *BSI* index for the femur ($F_{(1,23)} = 49.5$, $b = -0.347$, $p < 0.001$; Fig. 2c – continuous line) and humerus ($F_{(1,23)} = 55.2$, $b = -0.228$, $p < 0.001$; Fig. 1c – dashed line). We also detected a large effect size of body mass on the *ASI* index for the femur ($F_{(1,23)} = 54.7$, $b = -0.322$, $p < 0.001$; Fig. 2d – continuous line) and humerus ($F_{(1,23)} = 103.5$, $b = -0.291$, $p < 0.001$; Fig. 2d – dashed line).

## DISCUSSION

The ACI values for most pterosaur taxa investigated here are congruent with the hypothesis that these animals, including large and small forms, had a good terrestrial locomotion ability. The greater strength of the humerus relative to the femur also indicates that pterosaurs were generally forelimb dominant, with limbs supporting most of the body mass during terrestrial locomotion. Our results thus support the view of pterosaurs as forelimb-dominant quadrupeds, as previously suggested (Hwang et al. 2002, Chatterjee & Templin 2004).

However, our results indicate that some variation exists regarding forelimb-dominance in pterosaurs. For non-pterodactyloids, our results show high bending strength values for both stylopodia in all studied specimens, especially for the humerus of the two *Rhamphorhynchus muensteri* specimens. These results indicate a high strength of both stylopodia and suggest that in non-pterodactyloids forelimb-dominance was less pronounced than in pterodactyloids, wherein bending strength values show higher differences between humerus and femur (Table III).

These results may also support the reconstructions of the sprawling forelimbs of non-pterodactyloid pterosaurs. Wilson & Carrano (1999) argue that the more distant arrangement of the forelimbs relative to the centre of mass causes the humerus to be subject to a higher bending moment on its longitudinal-horizontal plane, thus, requiring

**Table III. The 25 specimens used and their respective values of mass (Kg), Bending Strength Index (BSI) and Axial Strength Index (ASI) for the humerus and femur. The axial bone length measurements, bone shaft diameter, shaft area, and section modulus can be accessed in Table SI.**

| Group | Species | Mass (kg) | BSI Hu (MPa^-1) | BSI Fe (MPa^-1) | ASI Hu (GPa^-1) | ASI Fe (GPa^-1) |
|---|---|---|---|---|---|---|
| Eudimorphodontoidea | *Arcticodactylus cromptonellus* (MGUH VP 3393) | 0,01 | 300,05 | 258,00 | 8,65 | 8,64 |
| Rhamphorhynchidae | *Rhamphorhynchus muensteri* (RAM 14522) | 0,52 | 85,12 | 21,98 | 1,73 | 0,87 |
| Rhamphorhynchidae | *Rhamphorhynchus muensteri* (SMNS 9620) | 0,68 | 46,89 | 20,95 | 1,24 | 0,83 |
| Rhamphorhynchidae | Rhamphorhynchidae indet. (NHMUK PV R 40126) | 0,59 | 175,69 | 64,63 | 4,26 | 2,56 |
| Anurognathidae | *Anurognathus ammoni* (Private specimen Bennett 2007) | 0,01 | 473,12 | 661,58 | 15,18 | 20,15 |
| Germanodactylidae | *Germanodactylus ramphastinus* (BSPG 1977 XIX1) | 0,35 | 134,75 | 116,53 | 3,64 | 3,30 |
| Ctenochasmatidae | *Pterodaustro guinazui* (MHIN-UNSL-GEO-V-241, hatchling) | 0,02 | 265,69 | 265,66 | 5,35 | 4,62 |
| Anhangueridae | *Anhanguera piscator* (NSM-PV 19892) | 22,70 | 63,34 | 13,75 | 1,08 | 0,51 |
| Anhangueridae | Anhangueridae indet. (BSP 1980 I 122) | 9,68 | 44,78 | 15,01 | 1,10 | 0,55 |
| Anhangueridae | Anhangueridae indet. (SMNK 1133) | 42,52 | 29,76 | 9,32 | 0,44 | 0,34 |
| Anhangueridae | *"Brasileodactylus sp."* (AMNH 24444) | 8,08 | 72,61 | 16,20 | 1,21 | 0,60 |
| Anhangueridae | *Anhanguera spielbergi* (RGM 401 880) | 47,96 | 37,17 | 6,10 | 0,65 | 0,26 |
| Anhangueridae | *Tropeognathus mesembrinus* (MN 6594-V) | 129,10 | 30,88 | 4,19 | 0,52 | 0,18 |
| Pteranodontidae | *Pteranodon longiceps* (YPM 1175) | 54,42 | 32,37 | 38,11 | 0,41 | 0,80 |
| Dsungaripteridae | Dsungaripteridae indet. (SHN.13) | 20,40 | 40,64 | 26,95 | 1,00 | 1,06 |
| Dsungaripteridae | *Noripterus complicidens* (IVPP RV 73001) | 1,51 | 81,34 | 23,31 | 2,08 | 1,28 |
| Azhdarchoidea indet. | *Bennettazhia oregonensis* (USNM 11925) | 10,25 | 77,97 | 50,54 | 1,64 | 1,96 |
| Tapejaridae | *"Nemicolopterus crypticus"* (IVPP V-14377, Sinopterinae indet., hatchling) | 0,02 | 374,04 | 333,15 | 7,06 | 10,41 |
| Tapejaridae | *Tupandactylus navigans* (GP/2E 9266) | 6,53 | 52,91 | 53,98 | 1,11 | 1,92 |
| Tapejaridae | Tapejara wellnhoferi (SMNK PAL 3985) | 1,36 | 51,98 | 23,88 | 1,23 | 0,85 |
| Azhdarchidae | *Cryodrakon boreas* (TMP 1992.83) | 29,87 | 47,57 | 33,22 | 0,75 | 1,17 |
| Azhdarchidae | *Hatzegopteryx thambema* (FGGUB R1083) | 202,52 | 124,32 | 23,92 | 1,48 | 0,85 |
| Azhdarchidae | *Quetzalcoatlus lawsoni* (TMM 47180) | 24,04 | 63,28 | 34,90 | 0,87 | 1,23 |
| Azhdarchiformes | *Montanazhdarcho minor* (MOR 691) | 10,36 | 48,44 | 50,08 | 0,84 | 1,76 |
| Azhdarchoidea indet. | Azhdarchidae? indet. (RBCM. EH.2009.019.0001) | 1,24 | 106,28 | 71,31 | 2,94 | 2,51 |

greater bone resistance to bending forces. Non-pterodactyloid pterosaurs, especially rhamphorhynchids and non-breviquartossans, have traditionally been reconstructed as having spread forelimbs (Wellnhofer 1970, Unwin 1988, Padian et al. 2004, Prondvai et al. 2012, Witton 2015), due to the lateral projection of the glenoid, and its dorsoventral asymmetry, allowing anterodorsal and posterodorsal movement of the humerus. However, these features restrict anterior and posterior movements by not allowing an abduction of the humerus below the scapula line (Witton 2015). This limb arrangement may limit the quadrupedal walking speed and leads to the hypothesis that these pterosaurs spent more time flying than moving on land, like seabirds (see Chatterjee & Templin 2004, Witton 2015). However, the morphology of the glenoid and humerus may vary among non-pterodactyloids. For example, the glenoid

of *Dimorphodon macronyx* resembles that of pterodactyloids, indicating a higher forelimb posture for these pterosaurs (Witton 2015). Despite supporting this postural conformation and overall bone strength, the exceptionally high values for the stylopodia of non-pterodactyloid pterosaurs may also be an effect of the negative relationship with body mass, as these taxa are all very small (body mass < 0.05 kg) (e.g., Morato 2006) in our study.

However, it must also be taken into account the evident influence of flight in the relative

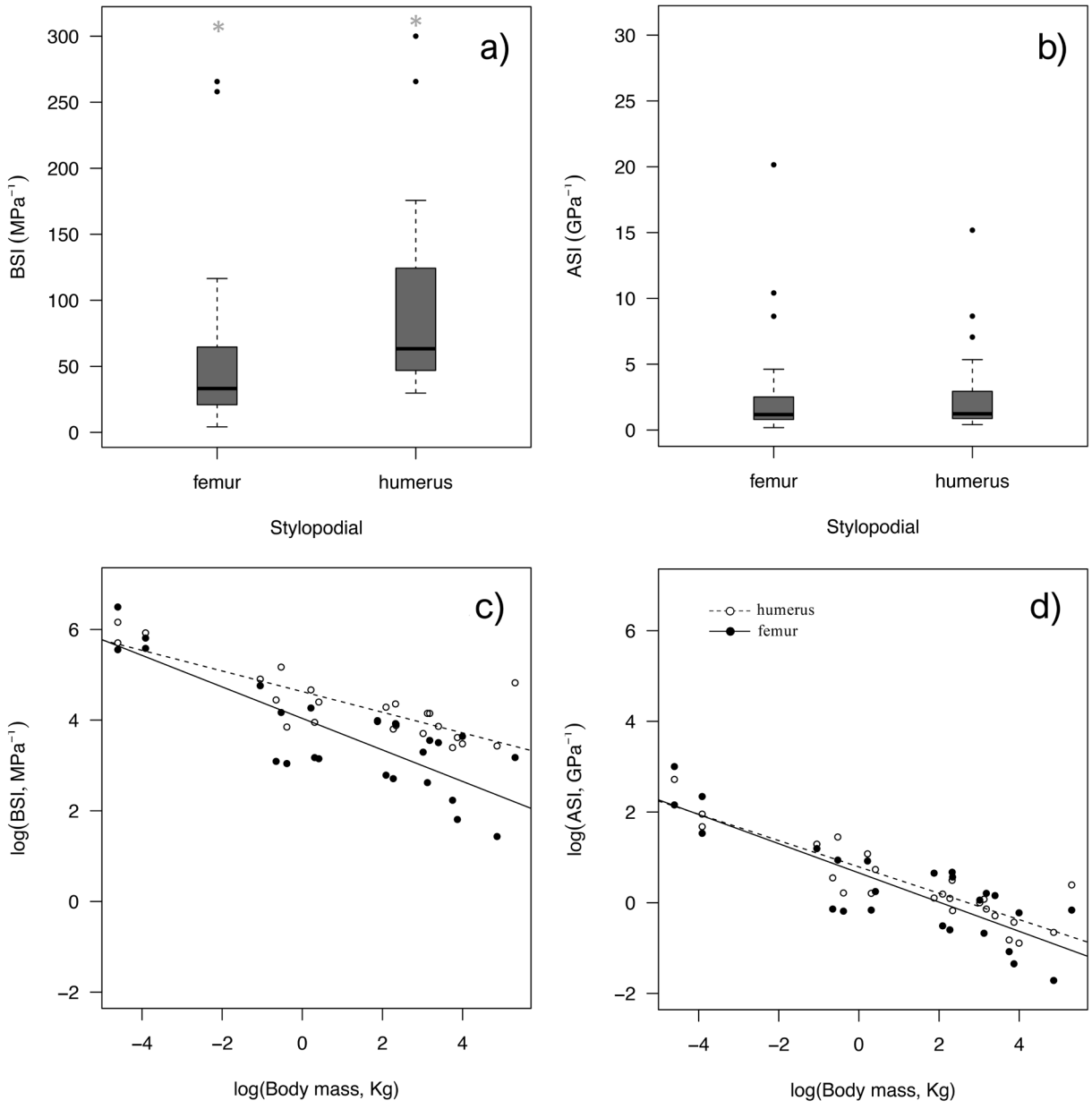



**Figure 2. Variation of the bending strength indicator (*BSI*) and the axial strength indicator (*ASI*) varied between the humerus and femur and their relationship with the body mass of pterosaurs. a) and b) Higher BSI and ASI indexes of the humerus than the n femur. c) and d) Negative relationships between the pterosaurs' body mass and *BSI* and *ASI* indexes for the femur (continuous lines) and humerus (dashed lines).**

strength indicators of the humerus. A significant portion of the observed difference in ACI between the femur and humerus in pterosaurs is likely attributable to the bending forces generated during flight. It is thus likely that the much greater variation in humeral ACI values (Figure 2, Table III) reflects the broad diversity of flight styles among the pterosaur species included in the sample. Indeed, diversity of flight styles in pterosaurs under the light of variation in humeral strength indicators has already been explored before (Witton & Habib 2010). Even though the present contribution aims at focussing on pterosaur terrestrial locomotion, exploring humeral ACI values under the light of flight performance/style could prove an interesting approach for future studies; as has been done for example for the giant cathartiform bird *Argentavis magnificens* (Vizcaíno & Fariña 1999).

According to Alexander (1983), relatively high ACI values suggest agility and a more remarkable ability to move relatively fast. The indexes recovered here for the humerus and femur are similar to those of modern quadruped animals that can move with good agility and even run at high speed (Alexander 1985, 1989, Biewener 1990, Fariña 1995, Casinos 1996, Bargo et al. 2000) such as the white rhinoceros (*Ceratotherium simum*) and the water buffalo (*Bubalus bubalis*) (Alexander & Pond 1992). The study by Chatterjee & Templin (2004) calculated speed estimates for some pterosaurs and concluded that large species could reach speeds over 30 km/h. Although not indicating speed estimates, our results suggest that pterosaurs generally possessed limb bones capable of withstanding the mechanical stress of running, to the exception of the giant azhdarchid herein analyzed, *Hatzegopteryx thambema* (Table III).

The relative robustness of the forelimbs of pterosaurs, especially pterodactyloids, has also led to the hypothesis that these animals could take off from quadrupedal launch using primarily the forelimbs for thrust (Habib 2008, Molnar 2009, Witton & Habib 2010, Benson et al. 2014, Witton 2015, Padian et al. 2021, Griffin et al. 2022, 2024). Our results support this hypothesis, as according to Alexander (1985), higher ACI values for a given limb indicate a greater capacity to support larger fractions of the animal's weight, so the humerus could help the load exerted during take-off.

Regarding the hypothesis that azhdarchoids generally possessed greater aptitude in ground locomotion (Witton 2008, Witton & Naish 2008, Padian et al. 2021), our results showed no significant difference among azhdarchoid humeri compared to other pterodactyloids. Our results indicate that humeral structural properties may be common between azhdarchoids and other pterodactyloids. Of course, this does not rule out the possibility that azhdarchoids could exhibit some variations regarding other biomechanical aspects (other than humeral ACI values) that could relate to a higher degree of terrestriality. Indeed, despite the similarity in humeral ACI values among pterodactyloids overall, femoral ACI values are actually higher in azhdarchoid species if compared to other pterodactyloid species; especially if compared among species of similar mass (Table III, *e.g. Tupandactylus navigans* vs. "*Brasileodactylus* sp."; *Montanazhdarcho minor* vs. BSP 1980 I 122; *Quetzalcoatlus lawsoni* vs. *Anhanguera piscator*). Interestingly, the same is true for dsungaripterids (Table III, compare SHN.13 and *Anhanguera piscator*). Humeral ACI values might be similarly high among pterodactyloids overall due to the influence of structural adaptations related to flight rather than terrestrial locomotion; while femoral ACI values, in turn, might be more intimately related to terrestrial locomotion than humeral ACI values. We propose, therefore, that

the observed variation could be related to a higher degree of terrestriality in azhdarchoids (or tapejaroids overall, including both azhdarchoids and dsungaripterids) if compared to other pterodactyloids, especially pteranodontoids. For a more comprehensive comparative picture among pterodactyloids overall, data on medium/large archaeopterodactyloids (such as some ctenochasmatids) are still needed.

The ACI values for the largest specimen of the study, the giant azhdarchid *Hatzegopteryx thambema*, were relatively low. These results do not contradict the hypothesis that azhdarchid pterosaurs possessed a remarkable ability to move on land (Witton 2008, 2015, Padian et al. 2021). However, they indicate that giant azhdarchids probably did not run, even though they could spend most of their time on the ground (Naish & Witton 2017, Padian et al. 2021).

## CONCLUSIONS

The approach used in this study was consistent with other biomechanical studies on pterosaur terrestrial locomotion. The values found are consistent with those of animals moving easily on land. The very high values found for non-pterodactyloid pterosaurs may be a limitation of the method when used on very small individuals, or indicate a high bending and compression resistance for their stylopodials. According to our results, all pterosaurs evaluated here share high ACI values, indicating that all possessed good resourcefulness in land locomotion. Nevertheless, pteranodontoids tend to exhibit lower femoral ACI values compared to similarly sized pterodactyloids, particularly azhdarchoids and dsungaripterids; corroborating previous hypotheses of azhdarchoids as more terrestrial forms if compared to pteranodontoids. Overall, athletic ability indexes can be a useful tool to infer terrestrial proficiency in pterosaurs as in other vertebrates.

Given our limited sample and the gaps in our data (with many values being estimations based on close taxa), this work is intended as an initial assessment that will hopefully foster further studies on pterosaur ACI. To provide a more comprehensive studies in the future, further data gathering will be needed. We encourage further studies, if possible and if preservation allows, to measure cortical bone thickness when describing new pterosaur fossil material.

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## SUPPLEMENTARY MATERIAL

**Table SI.**

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**Data availability**

Data will be made available upon reasonable request.

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DFC – Data Curation, Investigation, Methodology, Writing – original draft, Writing – review & editing. RVP - Conceptualization, Data Curation, Investigation, Methodology, Writing – original draft, Writing – review & editing. ASN - Data Curation, Investigation, Methodology, Writing – review & editing. AN – Methodology, Writing – review & editing. FRC – Conceptualization, Supervision, Resources, Writing – original draft, Writing – review & editing.

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